

Connectors and Cables -- Guided Notes

Unit tp-2.5 | CompTIA Network+ N10-009 Obj. FC0-U71 2.5 | N10-008 FC0-U71 2.5

Name: _____ Date: _____ Period: _____

Complete each question using the lesson reading. Write in the blank or answer in the space provided.

Question 1

USB Type-A connectors are the _____ (flat rectangular) end that plugs into the _____, such as a computer tower or charger. USB Type-B is a square-ish connector used on _____ such as printers. USB Type-C is reversible and used for both _____ and _____ in modern devices.

Question 2

HDMI cables carry both _____ and _____ signals in a single cable. HDMI 1.4 supports up to 4K at _____ Hz. HDMI 2.0 supports 4K at _____ Hz. DisplayPort can support higher _____ rates than HDMI, making it preferred for high-refresh gaming monitors.

Question 3

A Thunderbolt 3 or 4 port uses the same physical _____ connector as USB-C but supports speeds up to _____ Gbps, can carry _____ (video output), _____ (USB protocols), and even PCIe data. Thunderbolt is identified by the _____ icon next to the port.

Question 4

RJ-45 is the connector used for _____ networking cables (Ethernet). It has _____ pins. RJ-11 is used for _____ lines and has _____ pins. A _____ cable is used to connect a computer directly to another computer without a switch.

Question 5

3.5mm audio jacks are color coded: _____ (lime green) is for headphones/speakers (output), _____ (pink or red) is for microphone (input), and _____ (blue) is for line-in from an external audio source. The jack carries an _____ (digital/analog) audio signal.

Question 6

SATA data cables are _____ (wide/narrow) L-shaped connectors carrying only _____ between the motherboard and drive. SATA power cables come from the _____ and carry 3.3V, 5V, and 12V rails. Molex connectors are an older _____ (data/power) connector used for fans and older peripherals.

Question 7

The DB9 (DE-9) connector is a 9-pin serial port used for _____ devices such as older modems and network equipment console connections. The DB15 (DE-15) is a _____ pin VGA video connector. Both are examples of _____ -style connectors, named for their D-shaped metal shell.

Question 8 -- Real World Application

REAL WORLD: A student wants to connect a new 4K 144 Hz gaming monitor using a cable they found in a drawer. The cable is HDMI 1.4. They plug it in and the monitor only runs at 4K 30 Hz. Explain why this is happening, what the limitation is, and what cable they need to get the full 144 Hz at 4K.

Download the answer key from the class server:

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wget http://[SERVER]/resources/network-engineer/tp-2-5-connectors-and-cables/tp-2-5-connectors-and-cables-answer-key.pdf
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